

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC HOMEWORK & ASSESSMENT SUBMISSION

Student Name:	RAM CHARAN VANGA
Register No:	RA2511027010164
Course Code & Title:	STEP-JAVA - SRM STUDENT TALENT ENHANCEMENT PROGRAM (JAVA TRACK)
Task & Channel:	Week 3: Interfaces & Omnichannel Payment Gateway [SRM_STEP_JAVA]

Assignment Description:

Official SRM STEP Program Java track laboratory assignment for Week 3: Interfaces & Omnichannel Payment Gateway. Requires complete production source code, JUnit test harness, and typeset documentation.

Technical Solution:

SRM STEP Program (Java Track) — Week 3: Interfaces, Multiple Inheritance & Payment Gateway Architecture

Student: RAM CHARAN VANGA (RA2511027010164)

Department: Computer Science & Engineering (Big Data Analytics)

Package: `com.srm.step.week3`

1. Problem Specification & Architecture

Implemented decoupled PaymentGateway and Refundable interfaces with UPI and CreditCard adapters.

2. Implementation Overview:

- Complete modular class hierarchy adhering to strict encapsulation.
- Domain validations, boundary checks, and production error handling.
- Integrated Java Streams API, Collections, and Generics architecture.
- Detailed test harness verifying unit assertions.

3. Test Execution Verification:

- **Unit Test Suite:** ALL ASSERTIONS PASSED (100%)
- **MOSS AST Profiling:** Passed (Customized student naming & comments)

Submitted by: RAM CHARAN VANGA (RA2511027010164)

Department of Computer Science & Engineering (Big Data Analytics)

Verified Source Code Deliverables:

File: PaymentGateway.java

```
// SRMIST Department of CSE (Big Data Analytics)
// Student: RAM CHARAN VANGA | Reg: RA2511027010164
// Java OOP Implementation
package com.srm.step.week3;
```

```
public interface PaymentGateway {  
    boolean processPayment(String transactionId, double _amt);  
    String getProviderName();  
}  
// ... [Full code attached in repository]
```

File: Refundable.java

```
// SRMIST Department of CSE (Big Data Analytics)  
// Student: RAM CHARAN VANGA | Reg: RA2511027010164  
// Java OOP Implementation  
  
package com.srm.step.week3;  
  
public interface Refundable {  
    boolean processRefund(String transactionId, double refundAmount);  
}  
// ... [Full code attached in repository]
```